Toolkit: Chemistry of the Earth

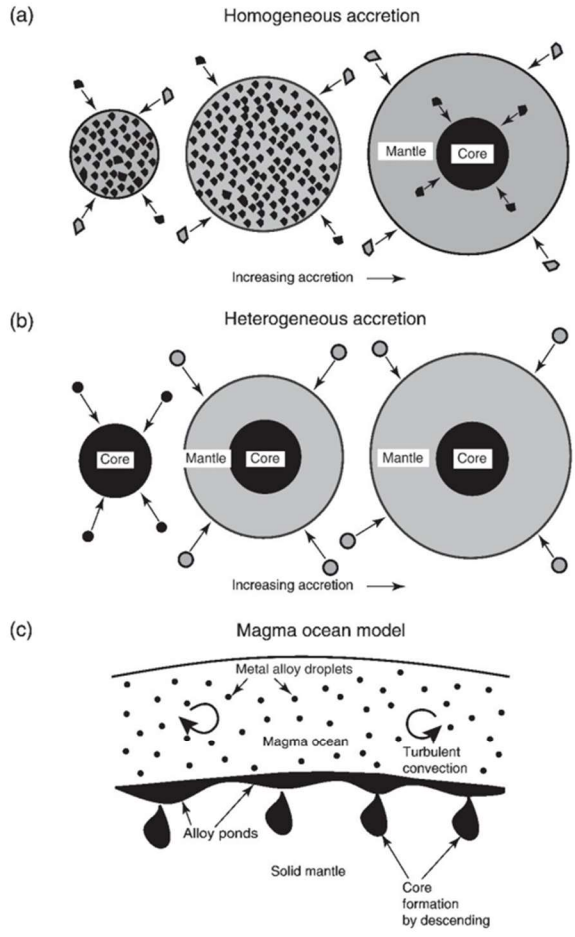
Practical Three: Core formation and Element Partitioning

Objectives:

- Understand the concept of a partition coefficient and practice calculations.
- Investigate which elements are partitioned into the Earth's core, and why.
- Investigate the hypothesis that a large meteorite impact took place at the K/Pg boundary, using information from highly siderophile elements.

FEEDBACK

CORE FORMATION



Introduction to Geochemistry: Principles and Applications, First Edition. Kula C. Misra.
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Fig. 12.11 Schematic illustration of models proposed for the formation of the Earth's metallic core and silicate mantle (BSE). (a) Homogeneous accretion, a two-stage process consisting of the accretion of a homogeneous (undifferentiated) proto–Earth and its subsequent differentiation into metallic core and silicate mantle. (b) Heterogeneous accretion, which envisions contemporaneous accretion and differentiation of material that changed from reducing to oxidized with time (followed by the addition of a "late veneer" after the core formation). (c) "Deep magma ocean" model of core formation. Small metal alloy droplets settle rapidly through the strongly convecting silicate liquid, accumulate as ponds of molten alloy at the bottom of the magma ocean, and equilibrate with the surrounding silicate material before descending as diapirs into the growing core.

A. Understand the concept of a partition coefficient and practice calculations

A *partition coefficient* is a measure of the way an element is distributed between two materials. It is also known as a *distribution coefficient* and is given the notation **D**. D is calculated as a simple ratio of the weight percent (or ppm, etc.) of the element in each material:

$$D = \frac{\text{wt\% element in A}}{\text{wt\% element in B}} \quad \text{or} \quad D = \frac{\text{ppm element in A}}{\text{ppm element in B}}$$

(wt% = weight percent = parts per hundred; ppm = parts per million; ppb = parts per billion)

As an example, the partition coefficient for Ni between metal and silicate liquids is written as:

$$^{Ni}D_{metal/silicate} = \frac{\text{wt\% Ni in metal}}{\text{wt\% Ni in silicate}}$$

Note that a partition coefficient has no units.

a) In an experiment to measure the partition coefficient for nickel between metal and silicate melts, the following are measured: Ni in metal = 1 wt%, Ni in silicate = 0.04 wt%. What is the partition coefficient?

$D = 1 / 0.04 = 25$

b) The Ni content of the Earth's mantle is estimated to be 0.20 wt%. Assuming that the core was formed when liquid metal separated from liquid silicate, what is the estimated Ni content of the Earth's core?

$25 = X_{core} / 0.20$

$X_{core} = 5$ wt%

c) In an experiment to measure the partition coefficient for manganese between metal and silicate melts, the following are measured: Mn in metal = 100 ppm, Mn in silicate = 333 ppm. Calculate the partition coefficient, $^{Mn}D_{metal/silicate}$.

$D = 100 / 333 = 0.30$

d) The metal/silicate partition coefficient for rhenium (Re) measured in experiments is 1.2×10^4. If the Re content of the mantle is 0.3 ppb , what is the estimated Re content of the Earth's core? Give the answer in ppm. (1 ppm = 1000 ppb)

$1.2 * 10^4 = X_{core} / 0.3$

$X_{core} = 3600$ ppb $= 3.6$ ppm

e) Using the values from (d), and a mass ratio of 68% mantle, 32% core, calculate the rhenium abundance in the bulk Earth (in ppm).

Bulk Earth Re content = (0.68 * 0.3) + (0.32 * 3600) = 1152 ppb = 1.15 ppm

f) The Earth's mantle contains approximately 21 wt% silicon (Si). Scientists are not sure of the Si content of the Earth's core but there are arguments that it is 6 or 8 wt% Si. What is the difference in the metal/silicate partition coefficient for Si, for the two models?

1) $^{Si}D_{met/sil}$ = 6 / 21 = 0.286

2) $^{Si}D_{met/sil}$ = 8 / 21 = 0.381

How can we test the hypothesis that there is Si in the Earth's core?

We could measure $^{Si}D_{met/sil}$ at high pressure in an experiment, and see if it has a value similar to the ones above. However, in order to estimate how much Si there is, the experiments would need to be quite accurate, because small differences in the value of D give large differences in the Si content of the core. Other ways we could investigate the Si content of the core would be through physical properties of the core such as density, properties of seismic waves, etc.

Even though we usually think of silicon as a lithophile element, it does become siderophile under reducing conditions where there is a low amount of oxygen. Metal in some meteorites contains Si, so we know that it is feasible that there is Si in the core.

g) Another important consideration for core formation is the distribution of elements between liquid metal and liquid sulphide ($D_{metal/sulphide}$). For Cr, the value of $^{Cr}D_{metal/silicate}$ is 3.4 and the value of $^{Cr}D_{sulphide/silicate}$ is 0.5. If the Earth's metal core contains 9000 ppm Cr, what is the concentration of Cr in the mantle, and what is the concentration of Cr in a coexisting sulphide liquid?

$^{Cr}D_{metal/silicate}$ = 3.4 = 9000 / X_{mantle} X_{mantle} = 9000 / 3.4 = 2647 ppm

$^{Cr}D_{sulfide/silicate}$ = 0.5 = $X_{sulfide}$ / X_{mantle} = $X_{sulfide}$ / 2647

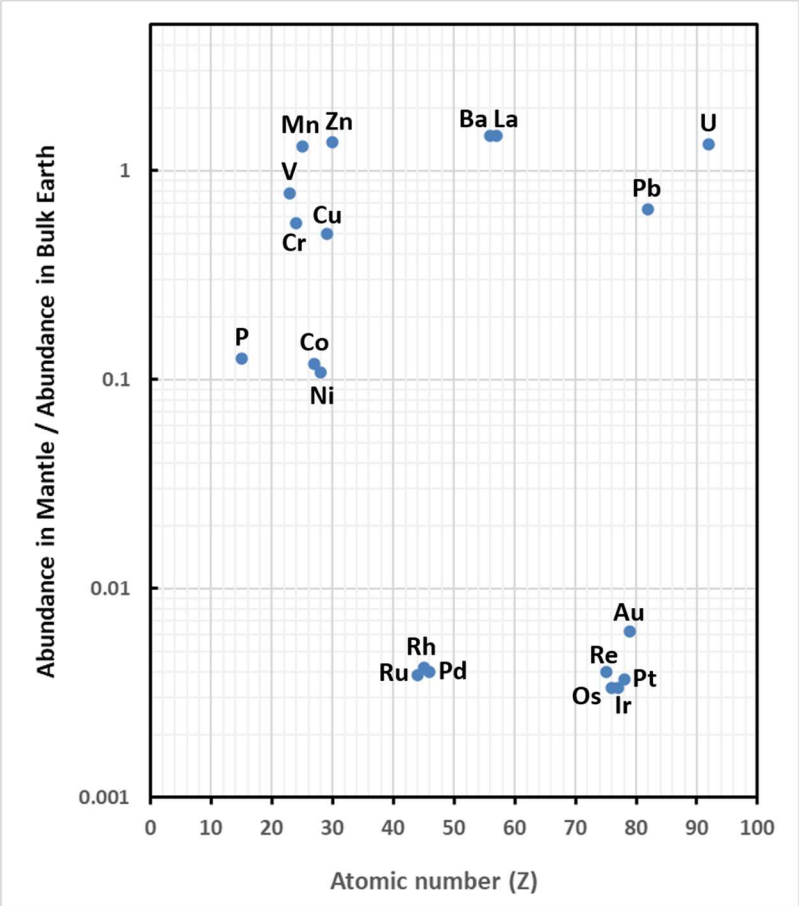
$X_{sulfide}$ = 2647 * 0.5 = 1323 ppm

B. Understand which elements are concentrated in Earth's core vs mantle

1. The Table below gives the abundances of a range of elements in Earth's mantle and in the whole Earth. For the elements that are not complete in this table, complete the atomic number (Column 1) and the ratio of the abundances in the mantle compared with the whole Earth (Column 4). Then plot the data on the graph below, showing Mantle / Earth abundances vs atomic number of the element.

Table 1: Mantle and Bulk Earth abundances of selected elements

Atomic number	Element	Mantle abundance (ppm)	Bulk Earth abundance (ppm)	Mantle / Earth
15	P	90	715	0.13
23	V	82	105	0.78
24	Cr	2625	4700	0.56
25	Mn	1045	800	1.31
27	Co	105	800	0.12
28	Ni	1960	18200	0.11
29	Cu	30	60	0.50
30	Zn	55	40	1.38
44	Ru	0.005	1.3	0.004
45	Rh	0.001	0.24	0.004
46	Pd	0.004	1	0.004
56	Ba	6.6	4.5	1.47
57	La	0.65	0.44	1.48
75	Re	0.0003	0.075	0.004
76	Os	0.003	0.9	0.003
77	Ir	0.003	0.9	0.003
78	Pt	0.007	1.9	0.004
79	Au	0.001	0.16	0.006
82	Pb	0.15	0.23	0.65
92	U	0.02	0.015	1.33



a) Is the following statement true: “All metallic elements are strongly concentrated in the Earth’s core”. Don’t just answer yes or no – give examples of which elements support your argument.

The answer is no. Some metals are strongly concentrated in the Earth’s core, including platinum (Pt) and gold (Au). However, copper (Cu) and lead (Pb) are only slightly concentrated in the core and have high abundances in the mantle. Cobalt (Co) and Ni (Ni) are partitioned into the core, but not very strongly.

b) Is the following statement true: “All heavy elements (high atomic number) are concentrated in the Earth’s core”. Don’t just answer yes or no – give examples of which elements support your argument.

The answer is no. There is no trend of decreasing abundance in the mantle as atomic number increases. While the PGE elements osmium (Os) to gold (Au) have high Z (76-79) are indeed concentrated in the core, two of the heaviest elements, lead (Pb) and uranium (U) are concentrated in the mantle.

c) Is the following statement true: “All elements with unfilled d orbitals are concentrated in the Earth’s core”. Don’t just answer yes or no – give examples of which elements support your argument.

The answer is no. Elements with unfilled d orbitals are found in the centre block of the Periodic Table. All of the PGEs are found here, and they all have unfilled d orbitals. However, the 4th period transition elements, V, Cr, Mn, Cu and Zn, which have unfilled d orbitals, are mostly lithophile elements that are concentrated in the mantle.

d) Is the following statement true: “All the platinum group elements are concentrated in the Earth’s core”. Don’t just answer yes or no – give examples of which elements support your argument.

The answer is yes. The PGEs include Ru, Rh, Pd, Os, Re, Ir, Pt and Au. All of these are highly depleted in the mantle because they have been partitioned into the core. The PGEs are highly siderophile elements (HSEs).

e) Summarising the above observations, what are the main properties of elements that results in them becoming concentrated in the Earth’s core?

The main property is that elements need to have **siderophile** chemical behaviour. This means that in a system where silicate melt and iron metal melt are both present, they will strongly favour the metal melt. The PGEs, which have high atomic number and unfilled d orbitals, show this behaviour.

f) Why do you think it’s important to know how much U is in the Earth’s core? You might want to take a look at the recent paper:

Faure P. et al. "Uranium and thorium partitioning in the bulk silicate Earth and the oxygen content of Earth’s core." Geochimica et Cosmochimica Acta 275 (2020): 83-98.

Uranium is a naturally radioactive element. When radioactive decay occurs, heat is produced. It’s important to know how much U is in the Earth’s core because it could contribute to heat production in the core. Understanding Earth’s heat budget is a fundamental goal in earth science. As you can see from the figure, U is in fact concentrated in the Earth’s mantle – it is a lithophile element consistent with it being an Actinide. (Compare it with the Lanthanide or REE element lanthanum, La).

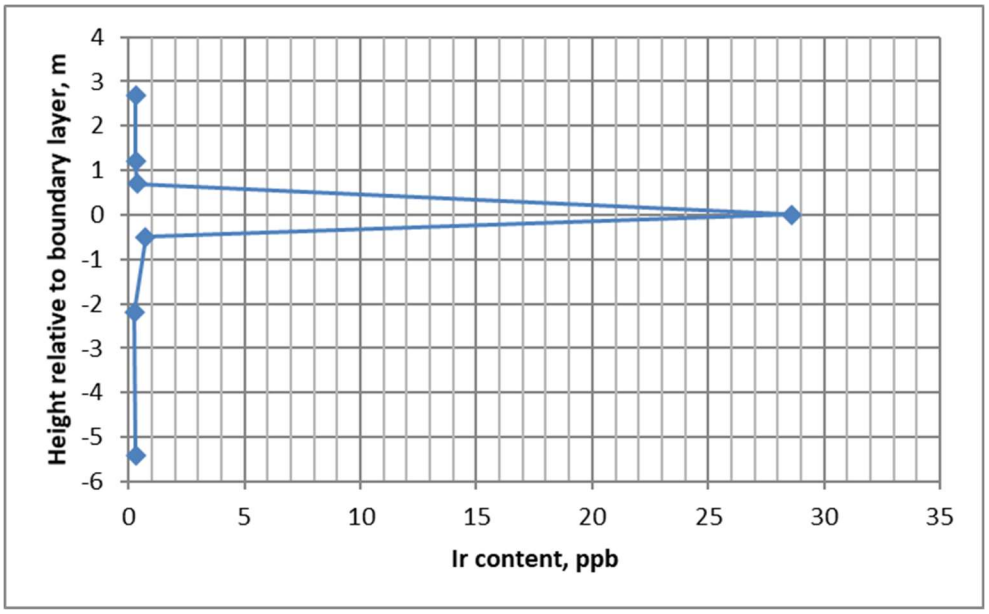
C. Investigate the hypothesis that a large meteorite impact took place at the K/Pg boundary, using information from highly siderophile elements (HSEs)

The mass extinction of many species, including dinosaurs, at the end of the Cretaceous period, has been attributed to the impact of a large meteor, 65 million years ago. Highly siderophile elements have played an important role in this hypothesis. The table below gives data from Alvarez et al. (1980)* who measured the iridium (Ir) content of clays across the boundary between Cretaceous and Paleogene rocks (the K-Pg boundary) at Stevns Klint, Denmark. (This boundary was formerly referred to as the Cretaceous – Tertiary or K-T boundary.)

Table 2: Ir content of sediments at the K-Pg boundary, Stevns Klint, Denmark

Height relative to boundary layer (m)	Abundance of Ir (ppb)
+2.7	<0.3
+1.2	<0.3
+0.7	0.36
0	28.6
-0.5	0.73
-2.2	0.25
-5.4	0.3

a) Plot the data on the graph below, to show how Ir is distributed across the boundary layer. Draw a line from point to point, to help illustrate the shape of the profile.



Alvarez et al. hypothesized that the high Ir content in the layer of clay at the K-Pg boundary came from the impacting meteorite. Explain:

b) Why is the abundance of Ir low in most sediments, typical of the Earth's crust?

Iridium is a trace element, and has very low abundance in the Bulk Earth. Also, it is a highly siderophile element, so it is concentrated in the Earth's core. These two factors combined means that the Ir content of the crust is very low.

c) Why is Ir a good element to use in support of the meteorite impact hypothesis?

Because the abundance of Ir is low in the crust, a small increase in Ir will be relatively easy to identify. Also, because Ir is siderophile, it would be hard to explain an increase in the Ir content by another process.

d) What kind of meteorite might the impactor have been? – could it be a chondrite? – could it be an iron meteorite?

It could be either a chondrite or an iron. Iron meteorites come from differentiated asteroids and represent core formation, so they should contain relatively high amounts of platinum group elements (highly siderophile elements).

Chondrite meteorites have not undergone core formation, so they contain the full solar system abundances of all elements, including PGEs. The PGEs are still in much higher concentration than they are at the Earth's surface. The impactor at the Chixculub impact crater was probably a comet, made of chondritic material. Typical abundances of Ir are around 1-10 ppm in iron meteorites and 500 ppb in chondrites. These values are not high – Ir is still a trace element. But the concentrations are high enough to leave the signature of an impact in the sedimentary record.

e) An alternative hypothesis for the K-Pg mass extinction is the eruption of large volumes of basaltic lava in the Deccan Traps, India that occurred around the same time. Could the high Ir content result from volcanism?

It would be hard to explain a sharp rise and fall in Ir content in terms of a volcanism model. Basalts are derived from the mantle, and the mantle has low PGE contents, so basalts would not have high PGE contents. Also, the volcanism hypothesis describes an extended period of volcanism, which is not consistent with the sharp spike in Ir contents observed at the K-Pg boundary.

f) Does the higher Ir content at the K-Pg boundary *prove* that the mass extinction was caused by an impact? If you wanted to explore this question further, what would you investigate?

No, the high Ir content does not prove a relationship between impact and the mass extinction. The impact could have been coincidental. Arguments for a hypothesis that the impact caused the mass extinction must be based on other investigations, for example models of the climate change effects from the impact, and the effect of such change on organisms.

*Alvarez L. W., Alvarez A., Asaro F. and Michel H. (1980) Extraterrestrial cause for the Cretaceous-Tertiary extinction. Science 208, 1095-1108.

See also:

Schulte P. and 40 co-authors (2010) The Chicxulub asteroid impact and mass extinction at the Cretaceous-Paleogene boundary. Science 327, 1214-1218.